How to install packages in Rstudio? (important)

1. **Installing packages from CRAN (Comprehensive R Archive Network).**

```
install.packages("ggplot2")
```

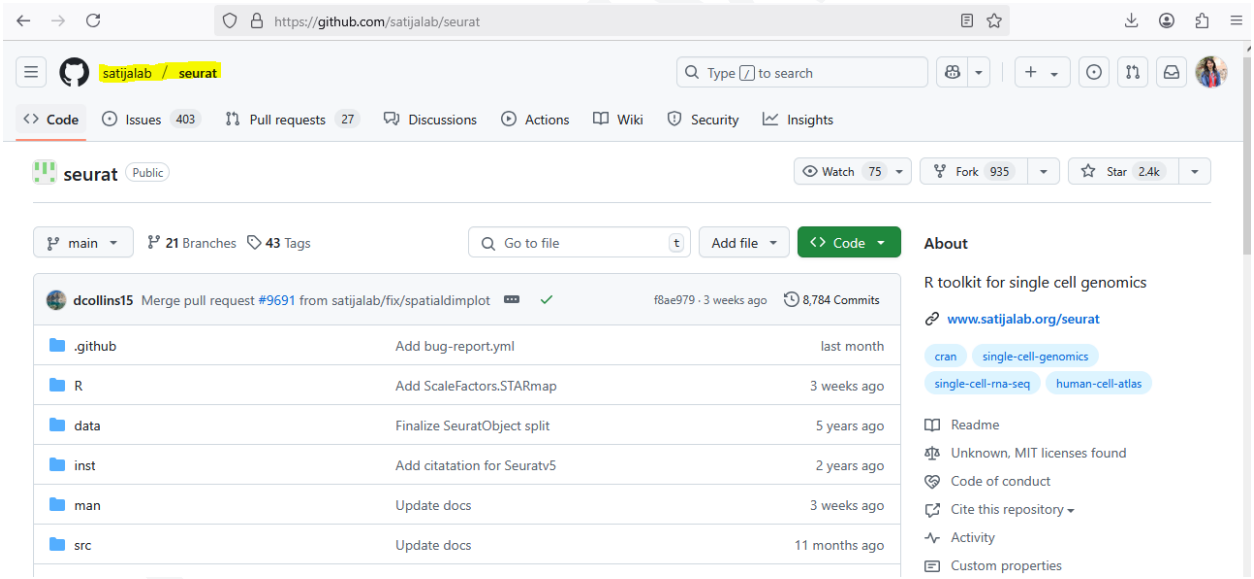
2. **Installing from Bioconductor**

```
install.packages("BiocManager")
BiocManager::install("DESeq2")
```

3. **Installing from GitHub using devtools or remotes**

```
install.packages("devtools")
# or
install.packages("remotes")

devtools::install_github("satijalab/seurat")
```



4. **Installing from a local file**

```
install.packages("/path/to/package.tar.gz", repos = NULL, type = "source")

install.packages("/path/to/package.zip", repos = NULL)
```

5. **Installing from a .Rprofile or R script**

```
if (!requireNamespace("ggplot2", quietly = TRUE)) {
  install.packages("ggplot2")
}
```

6. **Installing dependencies automatically**

Some packages have dependencies (other packages they rely on). You can ensure these dependencies are installed by setting the dependencies argument to TRUE when installing:

```
install.packages("ggplot2", dependencies = TRUE)
```

7. **Installing from a custom repository**

```
install.packages("package_name", repos = "http://myrepo.com")
```

Issues when installing packages in RStudio

1. **Missing dependencies**

   ```
   install.packages("package_name", dependencies = TRUE)
   ```

2. **Package compilation issues**

   ```
   sudo apt-get install build-essential
   ```

3. **Outdated R Version (most troublesome problem)**

   Certain packages may require a newer version of R than the one you have installed. This can result in errors when trying to install or load the package.

- **Solution**:
  - Check R version by running R.version.
  - If you need to update, visit the R Project website and download the latest version.

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- Some packages may only be compatible with the most recent version of R, so updating can resolve installation issues.

4. Package Version Incompatibility (most frequent problem)

   Consider installing a previous version of the package.

```
install.packages("remotes")
remotes::install_version("package_name", version = "x.y.z")
```

5. Permission Issues (on Windows/Linux/macOS)

install.packages("package_name", lib = "path_to_your_personal_library")

Or, to get root permission–

```
sudo R
install.packages("package_name")
```

6. Corrupted package or R library

Sometimes, R libraries become corrupted or packages don't get installed properly, causing errors when loading or using them.

```
remove.packages("package_name")

install.packages("package_name")
```

Alternatively, **clear the library** and reinstall all packages:

```
.libPaths()   # Check where your libraries are stored

unlink("/path_to_library", recursive = TRUE)   # Remove library folder
```

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